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RESEARCH ARTICLE

# Vegetation structure and climate shape mountain arthropod distributions across trophic levels

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## Abstract

1. Arthropods play a vital role in ecosystems; yet, their distributions remain poorly understood, particularly in mountainous regions.
2. This study delves into the modelling of the distribution of 31 foliar arthropod genera in the French Alps, using a comprehensive approach encompassing multi-trophic sampling, community DNA metabarcoding and random forest models.
3. The results underscore the significant importance of vegetation structure, such as herbaceous vegetation density, and forest density and heterogeneity, along with climate, in shaping the distributions of most arthropods. These responses to environmental gradients are consistent across trophic groups, with the exception of nectarivores, whose distributions are more sensitive to landscape structure and water availability.
4. By leveraging community DNA metabarcoding, this study sheds light on the understudied drivers of arthropod distributions, emphasizing the importance of modelling across diverse trophic groups to anticipate arthropod responses to global change.

KEYWORDS
bulk DNA metabarcoding, diets, French Alps, insects, random forest, species distribution model

## 1 | INTRODUCTION

Arthropods provide essential ecosystem functions, such as pollination, seed dispersal and nutrient recycling (Noriega et al., 2018; Weisser & Siemann, 2008). They also play a critical role in the stability of terrestrial food webs, benefiting various birds, mammals and reptiles (Noriega et al., 2018; Scudder, 2017). Yet, a growing number of studies show arthropods are declining globally due to human activities (Hallmann et al., 2017; Seibold et al., 2019; Wagner, 2020; Wagner et al., 2021). There is an urgent need to

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understand the environmental factors driving their range shifts and extinctions (Mammola et al., 2021; Suzuki-Ohno et al., 2020). However, the difficulty of sampling and identifying multiple arthropod taxa simultaneously has hindered the study of arthropod distributions across genera and trophic groups, especially at regional scale (Mammola et al., 2021). Hence, research investigating the main drivers of arthropod distributions tend to be biased towards charismatic or well-known taxa (e.g. Lepidoptera, Carneiro et al., 2014; Pellissier et al., 2013), to examine a small number of environmental predictors (e.g. McCabe & Cobb, 2021) or to be conducted at small scale (Hallmann et al., 2017; Hodkinson, 2005; Mammola et al., 2021). Besides, large-scale, multi-taxa studies often focus on measures of whole communities, such as biomass or diversity, rather than on the distributions of the taxa comprising these communities (e.g. Hallmann et al., 2017; Schuch et al., 2012).

Strong environmental gradients, for example, in mountainous areas, are unique models for identifying key drivers of arthropod distributions at regional scales, as they bring together very different environmental conditions in a small geographical area (Fukami & Wardle, 2005; Sundqvist et al., 2013). Moreover, the difficulties associated with regional-scale sampling and identification of arthropods can be overcome by combining multiple broad-spectrum complementary sampling techniques with recent advances in community DNA metabarcoding. The combined use of sweep netting, which samples arthropods from the herbaceous vegetation, and beating-cloth method, which are better suited to forests, allows representative sampling of foliar arthropods from the different types of vegetation stratified along elevational gradients (Cooper & Whitmore, 1990; Samu & Sárospataki, 1995). Community DNA metabarcoding allows for fairly standardized identification of sampled taxa without needing multiple experts for different arthropod groups, and overcomes the difficulty of identifying individuals or parts of individuals broken by sampling (Carew et al., 2018; Ji et al., 2013). Trophic databases facilitate the classification of sampled taxa into trophic groups (Le Guillarme et al., 2023; Rainford & Mayhew, 2015). However, such a combination of techniques has rarely been tested in the context of regional-scale multi-taxa and multi-trophic distribution models (but see Lim et al., 2022 for a study of community diversity along two elevational gradients).

In mountains, climate is the main driver of elevational vegetation stratification, but the location of vegetation types is also influenced by other environmental factors such as geological bedrock, soil properties, snow cover duration and topography. Thus, the use of multiple elevational gradients can help to partially disentangle climate and vegetation effects on arthropod distributions. Vegetation is expected to be an important driver of the distribution of foliar arthropods, in contrast to ground-dwelling arthropods, for which the relative importance of vegetation versus climate vary depending on taxa, trophic group and community metric (Uhey, Hofstetter, et al., 2020; Uhey, Riskas, et al., 2020). Indeed, foliar arthropods are closely associated with vegetation, using it as a habitat, a food source (e.g. herbivores such as Orthoptera) or a hunting ground (e.g. predators such as arachnids). The influence of vegetation is likely to be particularly strong for herbivores and nectarivores, given that most of them are specific plant specialists (Bernays & Graham, 1988; Tallamy & Shropshire, 2009) and that abrupt range limits are observed at the tree line for both trophic groups (e.g. skipper butterflies, Carneiro et al., 2014; or hemipterans, Mani, 2013). Vegetation can also influence higher trophic groups, for example, spiders needing web attachment sites. Among the various aspects of vegetation, especially vegetation structure, that is, density, average height and heterogeneity of herbaceous and tree strata, may be an important predictor, but direct tests are rare (Uhey, Hofstetter, et al., 2020).

Multiple elevational gradients also promise unique insights into how arthropod distributions are influenced by landscape structure, that is, by microhabitat heterogeneity and topographic structures in and around of the plot (Le Provost et al., 2021). In the ecotone between subalpine and alpine levels, patches of grasslands may explain the increase in grassland-related arthropods in the proximate forests. This, in turn, may favour predators in the ecotone by locally increasing the quantity and diversity of prey (Calderón-Sanou et al., 2022). While landscape-related factors are increasingly recognized as key drivers affecting arthropod biomass and diversity at large scale (Le Provost et al., 2021; Seibold et al., 2019; Suzuki-Ohno et al., 2020), this has been rarely explored in mountains and in the context of multi-taxa and multi-trophic distribution models.

Finally, the effect of soil variables on foliar arthropod distributions is even more rarely tested. Yet, soil pH, organic matter and water availability play significant roles in shaping the early developmental stages of soil-dwelling larval groups (Coulson & Whittaker, 1978; Zhao et al., 2023), and stone-content determines habitat availability for various above-ground arthropods (Cosandey et al., 2022; Mani, 2013).

In this paper, we combined sampling of 31 foliar arthropod genera belonging to five trophic groups with various local environmental predictors along 14 elevation gradients in the French Alps. We leveraged a machine learning algorithm capable of assessing the relative importance of predictors of arthropod distributions and accounting for non-linear responses and collinearity. We aimed to (1) test the reliability of distribution patterns obtained from metabarcoding data for key above-ground arthropod taxa, (2) quantify the relative importance of climate, vegetation structure, landscape structure, soil and water availability and (3) investigate whether the environmental response of foliar arthropods is consistent across trophic groups. Through these efforts, our research contributes to a deeper understanding of the factors shaping arthropod distributions at regional scale, particularly in mountain environments.

## 2 | MATERIALS AND METHODS

### 2.1 | Study sites

We studied the distribution of above-ground arthropods in the French Alps along 14 elevation gradients of the ORCHAMP long-term monitoring programme (Figure 1a,b, and Appendix 1). The

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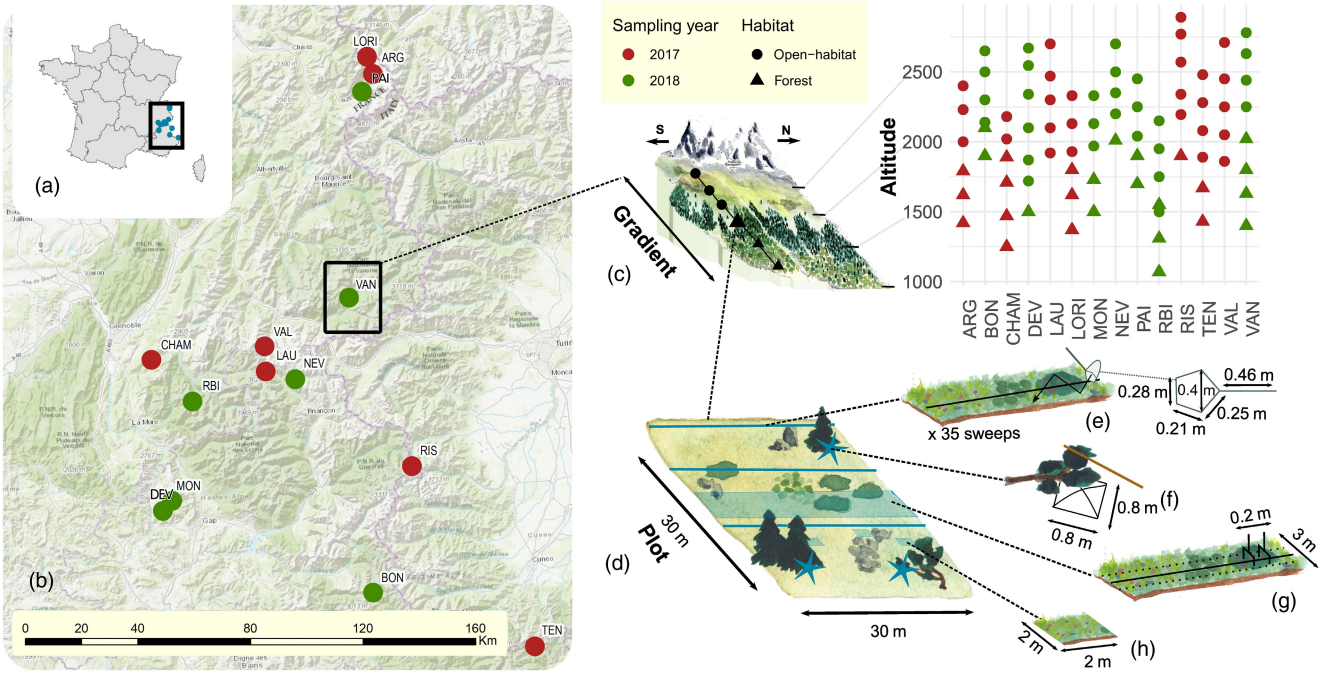



FIGURE 1 Study system. Location of the ORCHAMP observatory (a) and of the elevation gradients in the French Alps (b), example and actual repartition of the elevation and habitat per gradient (c), gradient names and year of sampling (b, c) and sampling design of one plot (d): randomly placed sweep netting transects (e), randomly chosen branches for beating-cloth method (f), permanent botanical sub-plot (g), permanent soil sub-plot (h).

gradients ranged from approximately 1000 to 3000 m and were selected to represent the environmental and topographic diversity found in the French Alps (Figure 1b,c). Each gradient featured consistent exposure and slope. We established a minimum of five 30 m x 30 m permanent sampling plots along each gradient, at 200 m elevation intervals. This resulted in a total of 83 plots, categorized into two broad habitats, 28 in forest and shrubland plots (called hereafter forest plots) and 55 non-forest plots (Figure 1c,d). Our study did not require ethical approval. Sampling was carried out with the authorization of the competent authorities (permits number 2018–221, 2018–241, 2018–379 and 2018–381, respectively by Vanoise, Mercantour, and Ecrins National Parks), or did not require authorization.

## 2.2 | Arthropod sampling and storage

We sampled arthropod communities along multiple elevation gradients over 4 weeks at the peak of the growing season, which minimized the effect of arthropod seasonality on our study (see Appendix 2, Figure S1 for precise sampling dates and Figure S2 for their independence from the model residuals). From the second week of July to the first week of August in 2017 and 2018, we used two protocols (Figure 1e,f). First, in all habitats, three 30 m long transects were sampled orthogonally to the slope with a sweep net with a 0.46 m handle and of ca. 0.1 $m^2$ aperture (ca. 35 sweeps, Figure 1d,e). Second, in forest plots, we additionally collected branch-dwelling arthropods by beating two times a low branch from three randomly selected trees with an 80 × 80 cm beating cloth underneath (Figure 1d,f). This sampling resulted in three bulk samples per non-forest plot, and six bulk samples per forest plot. Bulk arthropods were killed and preserved in 100% ethanol for 24 h, enabling rapid dehydration of the samples and protection of their DNA from enzymatic and chemical breakdown (Quicke et al., 1999). After 24 h, we drained the ethanol and then dried the samples using silica gel. Drying samples after ethanol pretreatment has been shown to improve DNA yield for a variety of organisms (Akinnagbe et al., 2011; Ballare et al., 2019). To standardize the number of samples across all plots, each bulk sample obtained by sweep netting was combined with one of the bulk samples acquired through beating in the forest areas. In total, we obtained 249 samples (i.e. three repetitions, 83 plots) from the 14 elevation gradients, 165 in non-forest plots and 84 in forest plots.

## 2.3 | Community DNA metabarcoding of the bulk samples

We performed community DNA metabarcoding analyses of the arthropod bulk samples, meaning that all individuals from a single sample were analysed simultaneously. This cutting-edge technique has gained popularity in environmental monitoring due to its ability to rapidly identify a wide range of taxa (Braukmann et al., 2019; Carew et al., 2018; Ji et al., 2013). We extracted intracellular DNA from each sample using the DNeasy Blood and Tissue Kit (QIAgen GmbH), following the manufacturer's instructions but adapting the

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volumes of lysis buffer to the volume of arthropods in each bulk sample and initially incubating the samples overnight with gentle agitation. We kept only a 125 $mm^3$ piece of the largest specimens' abdomens, to reduce their potentially disproportionate contribution to reads abundance (for more details see Appendix 3). We used the 'fwhF2' and 'fwhR2n' primer set to target a short fragment of the cytochrome c oxidase 1 (CO1) region, suitable to amplify arthropod DNA (Elbrecht et al., 2019; Vamos et al., 2017). To distinguish individual PCR replicates after sequencing, we added unique eight-base tags to the 5′ end of each primer. PCR reactions were replicated four times for each DNA extract sample along with appropriate controls (Appendix 3). Library construction followed the METAFAST protocol (in-house protocol developed for LECA, Genesupport accredited ISO-17025:STS 0680 pipeline for Life Science Fasteris NGS services). NextSeq pair-end sequencing technology was used to generate the sequence data on the Illumina platform at FASTERIS, Geneva, Switzerland. For more details, see Appendix 3.

## 2.4 | Bioinformatic curation pipeline

We first pre-processed the sequencing data using the OBITools software (Boyer et al., 2016). Forward and reverse paired-end reads were assembled based on their overlapping 3′-end sequences, demultiplexed, dereplicated and assigned to their original PCR replicate based on their unique eight-base tags. We extracted reference sequences for the CO1 region from the EMBL database version 142 by in silico PCR (ecoPCR, Ficetola et al., 2010) using the 'fwhF2' and 'fwhR2n' primer sequences (Elbrecht et al., 2019; Vamos et al., 2017, Appendix 3, 1), and allowing for three mismatches in both forward and reverse primer matches. Taxonomy was assigned to the sequences based on these reference sequences using the ecotag program from the OBITools.

Second, we used the R package metabaR (Zinger et al., 2021) to identify and remove tag-jumps, sequencing, extraction and PCR contaminants, dysfunctional PCR replicates and PCR errors such as chimera sequences. Sequences with taxonomic annotations <80% identity or not assigned to arthropods were also removed. We then excluded samples with two or fewer remaining PCR replicates, and averaged the PCR replicates per sample, leaving 244 retained samples at this stage. Quality checks and further details of the curation pipeline are provided in the Appendix 4.

Next, we clustered the curated sequences into Molecular Operational Taxonomic Units (MOTU) using SUMACLUST (Mercier et al., 2013), with a clustering threshold of 94% similarity (Bonin et al., 2023). The taxonomic annotation of the cluster centre was kept for each MOTU. To ensure consistency in the analysis, genera not recorded in France were removed using the GBIF filter tool (2.8% of genera removed, https://github.com/nleguillarme/gbif-filter-python). We then rarefied the sequence read counts without replacement and extrapolated to the first quartile for small samples to account for sampling depth differences.

Finally, to enable community diversity analyses, we first standardized MOTUs read counts by the total number of reads in each sample, then averaged these standardized MOTUs abundances at the plot level, resulting in a community matrix of MOTUs per plot (3632 MOTUs, 83 plots). Then, to enable individual taxa distribution models, we aggregated the MOTUs at the genus level and only considered the genera presence–absence, resulting in a community matrix of genera per plot (663 genera).

## 2.5 | Assignment to trophic groups

We assigned MOTUs and genera to one or more trophic groups using a literature search. We first used Rainford and Mayhew (2015), who inferred trophic groups for all Hexapoda at family level, as a baseline reference for Hexapoda, and the references within Le Guillarme et al. (2023) for other arthropod groups (Appendix 5 and Figure S3). When we were able to find information at genus level for the modelled genera, we corrected the trophic groups initially assigned at higher taxonomic level using the references in Appendix 5 and Tables S1 and S2.

## 2.6 | Climatic data

In mountain ecosystems, the presence of snow determines the onset and length of the growing season. We expected heat accumulation (i.e. energy available for growth) and frost exposure (i.e. physiological limit of most metabolic processes) during the growing season to shape the distribution of arthropod genera. In each plot, we thus calculated Growing Degree Days (*GDD*) to characterize heat accumulation, as the sum of the average daily degrees above zero accumulated over the growing season each year, averaged over 10 years (i.e. 2009–2018), and modelled in the first soil horizon (to 10 cm depth). We also considered the cumulative solar radiation over each growing season, averaged over the period 2009–2018 (*Solar radiations*). We performed the calculation of the GDD and solar radiations from the SAFRAN-SURFEX/ISBA-Crocus-MEPRA reanalysis, a model that deals with weather and snow conditions in mountainous regions based on massifs, elevation in 300 m bands, slope and aspect (Martinez-Almoyna et al., 2020; Vernay et al., 2022; Weil et al., 2021). We also computed the annual temperature range as the standard deviation of the monthly mean temperatures*100 (*T. seasonality*), as well as the average temperature of the three coldest months of the year to characterize frost exposure (*Average T. cold quarter*). For the latter two variables, we used the *biovars* function in the *R* package *dismo* (Hijmans et al., 2017) with data from in situ sensors permanently recording soil temperature in the centre of ORCHAMP plots between 2016 and 2018 (https://doi.org/10.57745/TVAHUQ).

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## 2.7 | Vegetation data

To study the structure of the herbaceous stratum, expert botanists recorded plant species using pinpoint sampling along a 30-m-long transect, positioned orthogonally to the slope at the centre of the plot. At 20 cm intervals along the transect, botanists performed two measurements: one 25 cm upslope and the other 25 cm downslope from the transect (Figure 1g). At each 20 cm increment, they identified all individuals that came into contact with the pins positioned at the two measurement points. This resulted in 300 pinpoint positions per plot (each with potentially multiple plant individual contacts). To measure the density of the herbaceous stratum, we calculated the total number of contacts for each plot (*Veg. density*). To characterize the dominant functional strategy of each plot, we computed the community weighted mean of plant height (*CWM Height*) and of specific leaf area (*CWM SLA*). We used species-level mean traits from our own measurements, which covered over 85% of the recorded species for the two traits. Any missing height values was supplemented using either TRY data (<1% of missing values, Kattge et al., 2020), expert estimations (2.5% of missing values) or height values from sister species (2% of missing values). Missing SLA values were also completed using TRY data (<1% of missing values), or were inferred with the *mice* function using height information (13% of SLA values, van Buuren & Groothuis-Oudshoorn, 2011).

In addition, we characterized the structure of the tree stratum in the forest plots. We measured the diameter of each tree within a 1.5-m distance to the botanical transect. For large trees only (diameter >30 cm), this protocol was extended to cover the entire 30 m × 30 m plot. This allowed us to characterize forest density by calculating both the sum of the basal area of small (diameter <7.5 cm) and of large trees per square meter (*Surf. of small*—respectively *large*—*trees*). We also characterized the heterogeneity of the forest structure by applying the Gini coefficient to all tree trunks' diameter classes combined (*Forest struct. heterogeneity*, see Bourdier et al., 2016 for an application on tree diameters; Gini, 1921). A Gini coefficient of zero indicates identical tree diameters, reflecting a stand of the same age, while a maximum value of 1 signifies a highly heterogeneous forest with maximum differentiation between tree basal areas (Sterba, 2008).

## 2.8 | Soil data

We sampled soil in the first soil horizon (0–10 cm) in three 2 m × 2 m subplots located 5 m below the botanical transect in each plot (Figure 1h). We measured pH (*Soil pH*) and soil organic matter (*SOM*) after 2-mm sieving (see Martinez-Almoyna et al., 2020) and averaged the three measures in each plot. In addition, we dug a soil profile on each plot, described the soil and sampled the morphological horizons. The weight content (%) of coarse elements (*Coarse elements content*) in the topsoil (0–10 cm) was obtained after air-drying, sieving to 2 mm and weighing the fraction >2 mm (coarse fragments). Where the topsoil consisted of several morphological horizons, we computed the coarse element content as a weighted average of the value of each horizon, weighted by its relative thickness.

## 2.9 | Landscape data

We used the OSO land cover map at 10 m to compute the relative cover of different landscape components in a 100 m radius around each plot (Inglada et al., 2017). We considered both anthropogenic elements (built-up areas) and natural elements such as forests (broad-leaved, needle-leaved), grasslands (meadows), shrublands (heath), mineral surfaces and water bodies. We computed the Shannon entropy diversity index of the relative proportion of landscape elements as an indicator of landscape heterogeneity (*Landcover heterogeneity*).

Additionally, we collected aerial images from the IGN BDOrtho database (https://geoservices.ign.fr/bdortho), which provides colour (RGB) and near-infrared (IR) reflectance at 20 cm resolution. For each plot, we extracted images at 30 m and at 100 m to cover the heterogeneity within plots and in their vicinity, respectively. Following the spectral species algorithm (Féret & Asner, 2014), we used the k-means clustering algorithm at the pixel-level for all images pooled together (Drake, 2013). The goal was to identify spectral clusters representing visual patterns in the images that are consistently observed across all plot landscapes. For each plot, we then computed the relative cover of different spectral clusters and derived spectral heterogeneity using the Shannon index within and around the plot (*Plot*—respectively *Plot vici.*—*spectral heterogeneity*).

## 2.10 | Water availability data

We approximated water availability in each plot using indicators of water in the plant community, in the soil and by the proximity of watercourses. We thus calculated the CWM of the Landolt moisture value of the herbaceous stratum (*CWM Moisture*). We ignored any missing data, as it only concerned rare species (12% of species, but >10% abundance only for one plot). We obtained topsoil water retention capacity (*TWRC*) using the coarse element content and the water retention capacities, estimated by pedotransfer functions, at field capacity and at the wilting point. We chose the pedotransfer equation corresponding to the best model for topsoil in Dobarco et al. (2019), which uses organic matter, clay and sand content. Finally, we computed the proximity of watercourses (*Prox. of waterc.*), which represents the area covered by watercourses within a 100 m radius, using the BD TOPAGE 'cours d'eau' shapefile (https://catalogue.datara.gouv.fr/geonetwork/srv/api/records/dd65bcf5-0a52-4d78-90ec-9ff4c5d76b15).

## 2.11 | Statistical analyses

We modelled the distribution (i.e. probability of presence) of the 38 arthropod genera that we found in at least 10 plots based on the above-described sets of predictors (i.e. four climatic, six vegetation, four soil and three water predictors), but then only kept the 31

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genera models with convincing performance (see results). Missing temperature seasonality (4%), average temperature of the four coldest months of the year (4%), soil water storage capacity (8%) and stone content values (8%) were inferred using the *mice* and *quickpred* functions (van Buuren & Groothuis-Oudshoorn, 2011). All predictors were centred and scaled.

We used random forests, a machine learning method particularly efficient when non-linear relationships and interactions between predictors are expected. Because some predictors were correlated (Appendix 6) and given our reduced sample size, we used a conditional recursive partitioning, which has been shown to be unbiased (*cforest_unbiased* method of the R *party* package, Hothorn et al., 2006). We trained a separate model for each of the 38 genera. The number of trees (*ntree*) was set to 2000, a good compromise between computing time and stability of the results (Probst et al., 2018). For each genus/model, we also determined the optimal number of variables randomly sampled at each split (*mtry*) using the *train* function of the R *caret* package (Kuhn, 2008). We used the *train* function to find the *mtry* value that maximizes the true skill statistic (TSS) of the model following the one-SE rule (Breiman et al., 1984) within a repeated cross-validation (10-folds, five replications). The one-SE rule selects the simplest model (in this case, the smallest *mtry*) that is within one standard error of the best-performing model, striking a balance between model complexity and performance. To evaluate the final model for each genus, we calculated the TSS on the whole dataset. We used the TSS because it has the advantage of accounting for both the sensitivity and specificity of the model, without being overly sensitive to taxa prevalence (Allouche et al., 2006).

Finally, we estimated the conditional importance of predictors as the average decrease in accuracy using the *varimp* function (Strobl et al., 2008).

For comparison with more classical multi-trophic studies, we also performed $\alpha$- and $\beta$-diversity analyses of the MOTUs at the community level. See Appendix 6 for more details on the methods and results of those supplementary analyses.

## 3 | RESULTS

### 3.1 | Reliable distribution patterns from metabarcoding data for key above-ground arthropod taxa

In general, our distribution models worked relatively well for all genera with an average TSS of 0.65 for the cross-validation and 0.82 for the final, optimized, models. All genus distribution models reached a minimum TSS of 0.2 and 82% (31 genera) reached a TSS higher than 0.5 for the cross-validation part, which we considered as reliable models (Figure 2a). These 31 genera belonged to different trophic groups, most often with mixed regimes, including parasitism, predation, herbivory, nectarivory and detritivory. The performances of the models tended to decrease with higher trophic levels, that is, from the detritivorous, herbivorous, nectarivorous, predatory to parasitic genera (Figure 2b). In the following, we consider only the 31 genera for which the predictive models were of sufficient quality.

### 3.2 | Vegetation structure and climate drive above-ground arthropod distributions

The distributions of the 31 genera were mainly influenced by vegetation (six out of 10 most important variables) and climate (three out of 10 most important variables, Figure 3). The most important predictors in these predictor group were vegetation density, a proxy for tree density (surf. of large trees) and the heterogeneity of forest structure. A proxy for water availability (CWM moisture), landcover heterogeneity and spectral heterogeneity in and around the plot played a lesser role, while the relative importance of temperature seasonality and other predictors linked to the presence of water in the soil or landscape and to the soil physico-chemistry was often negligible (Figure 3). For comparison, the $\alpha$-diversity of the whole community followed a similar trend, with over 80% of the relative importance of variables held by vegetation variables and GDD (Figure S4).

### 3.3 | Few trophic group-specific responses to the environment

Variable importance for distributions was mostly consistent across genera of all trophic groups. Except for nectarivores, vegetation structure and climate were predominant, which is consistent with the general pattern across the 31 retained arthropods genera (Figure 4). Specifically, most herbivores (e.g. *Chorthippus*, *Omocestus* and *Nithecus*) and two genera with a mixed diet between herbivory and predation (*Tenthredo* and *Metrioptera*) tended to show strong and positive responses to predictors related to the structure of the herbaceous stratum, particularly its density (Veg. density, Figure 5). Most strict predators and detritivorous and/or microbivorous groups showed positive responses to forest density (e.g. *Philodromus*, *Mitopus*, *Platypalpus*, *Stenopsocus* and *Entomobrya*, Figure 5, Surf. of large trees), or to the forest structure heterogeneity (e.g. *Hemerobius*, Figure 5).

In contrast, the distributions of nectarivores were mainly structured by water availability, that is, CWM moisture (e.g. *Scythris* responded negatively, *Eusphalerum* positively, Figures 4a and 5), TWRC (e.g. *Ematurga* responded positively, Figure 5) or to the proximity of watercourses (e.g. *Helina* responded positively, Figure 4a). Additionally, the distributions of nectarivores and those of arthropod genera with a combination of nectarivory and predation in their diet, as well as some strict predators, exhibited a positive correlation with variables related to landscape heterogeneity at both the plot scale (e.g. *Malthodes* and *Rhagonycha*, Figure 5, Plot spectral heterogeneity) and in the vicinity of the plots (e.g. *Eusphalerum*, Figure 5, Landcover heterogeneity).

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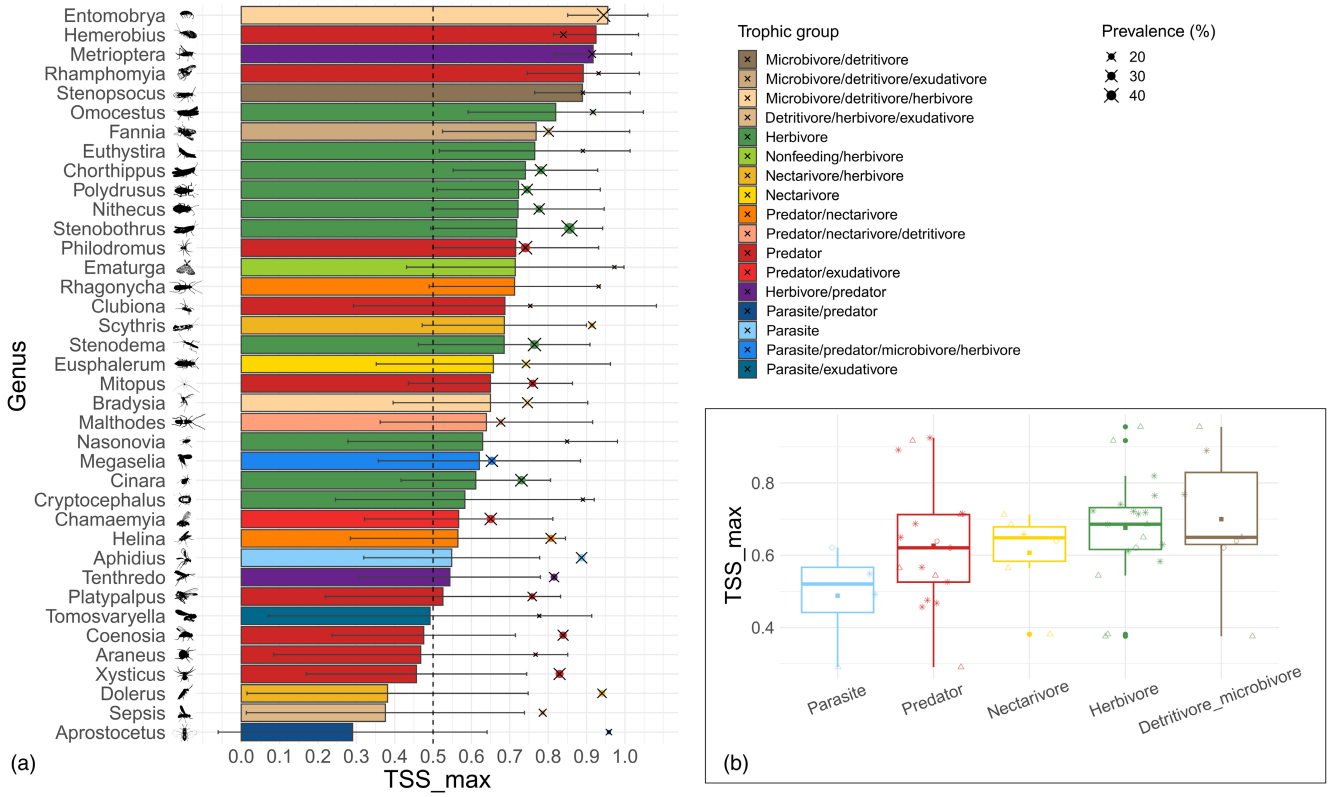



FIGURE 2 (a) Predictive quality of the different distribution models for the different genera. (a) Barplot representing the estimated true skill statistic (TSS) through cross-validation. The cross on each barplot represents the TSS of the final optimized models (X) for each modelled arthropod genus. The size of the circle represents the prevalence. The colour of the barplot represents the trophic group. (b) Average cross-validated TSS for parasites, predators, nectarivores, herbivores and detritivores-microbivores. Filled squares represent the mean. The other shapes correspond to the number of trophic groups, with genera belonging to several trophic groups represented in each corresponding boxplot (one trophic group for stars, two for triangles, three for circles and four for diamonds).

A comparable relative variable importance was observed in the models of α-diversity of the main trophic groups (Figure S5), with some minor differences detailed in the Appendix 6.

## 4 | DISCUSSION

### 4.1 | Reliable distribution patterns from community DNA metabarcoding data for key above-ground arthropod taxa

In our study, the combination of broad-spectrum sampling techniques, community DNA metabarcoding and random forest models allowed to produce arthropod genera distribution models of satisfactory quality (31 genera correctly modelled over 38, Figure 2), and to reveal biologically meaningful drivers of arthropod distributions at regional scale (Figures 3 and 4). We used sweep netting and beat clothing, which allowed our study to well represent foliar arthropods from different trophic groups and orders (particularly, Orthoptera, Diptera, Hemiptera and Hymenoptera, Figure S3). As a consequence, ground dwelling arthropods such as Coleoptera, Collembola or Formicidae were under-represented. To enhance model representativeness and generalizability across all arthropods, future efforts could incorporate sampling techniques specifically targeting organisms living on the soil surface (e.g. pitfalls traps, Pentinsaari et al., 2020; Samu & Sárospataki, 1995). Moreover, our study shows that community DNA metabarcoding offers promising avenues for generating insights into arthropod distribution shifts in response to global change on a larger scale, covering a wider range of taxa, and achieving faster and more-effective results compared to traditional methods (Ji et al., 2013; Pentinsaari et al., 2020), which are often limited to certain well-known taxonomic groups or indicator species (Hallmann et al., 2017; Ji et al., 2013). However, some limits remained. For example, the current quality of taxonomic assignments did not allow us to work at species level, resulting in a loss of resolution in the studied distributions. Future studies would likely benefit from improvements in the accuracy of taxonomic assignment of DNA sequences, for example, by using longer segments of the CO1 marker during amplification. Currently, this option is still constrained by the need to find highly conserved primers to avoid amplification biases and the quality of the DNA sequences extracted (Braukmann et al., 2019). Additionally, the accuracy of taxonomic assignments of DNA sequences improves with increased coverage (i.e. number of taxa and phylogenetic diversity of the database) and the depth of reference databases (i.e. number of referenced sequences by taxa, Andújar et al., 2018; Somervuo et al., 2017). While the CO1 gene's reference databases are already extensive for

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**FIGURE 3** General variable importance across all arthropod distributions. Estimated relative variable importance from random forest models to explain all 31 arthropod genera exhibiting an average cross-validation TSS higher than 0.50. The variable importance was adjusted based on the average cross-validation TSS of the respective model. The colours correspond to the type of the environmental variable.

arthropods (Andújar et al., 2018), the comprehensiveness and accessibility of these databases remain pivotal for advancing distribution models based on metabarcoding data at a macro-ecological scale for arthropods and other taxa (Somervuo et al., 2017).

## 4.2 | Vegetation structure and climate shape arthropod distributions

Our study underscored the significant and general influence of vegetation structure and climate on arthropod genera distributions, while revealing a more minor impact of landscape heterogeneity and soil physico-chemistry variables (Figures 3 and 4). This observation, also echoed by diversity models (Figures S5 and S6), represents new information about foliar arthropod communities along elevation gradients in mountains.

Vegetation structure emerged as a crucial factor in shaping the distribution of many arthropod genera, with, depending on the focal genus, either forest structure or herbaceous layer structure dominating (Figure 4). As a result, the compositions of forest and grassland differed in our system (Figure S6), which is consistent with earlier publications (Carneiro et al., 2014; Lasmar et al., 2020). On one hand, forests with numerous large and old trees particularly favoured certain arthropod genera (Figures 4 and 5, *Surf. of big trees*), potentially because old-growth forests provide a higher number of micro-habitats (e.g. conks of fungi, bark feature and

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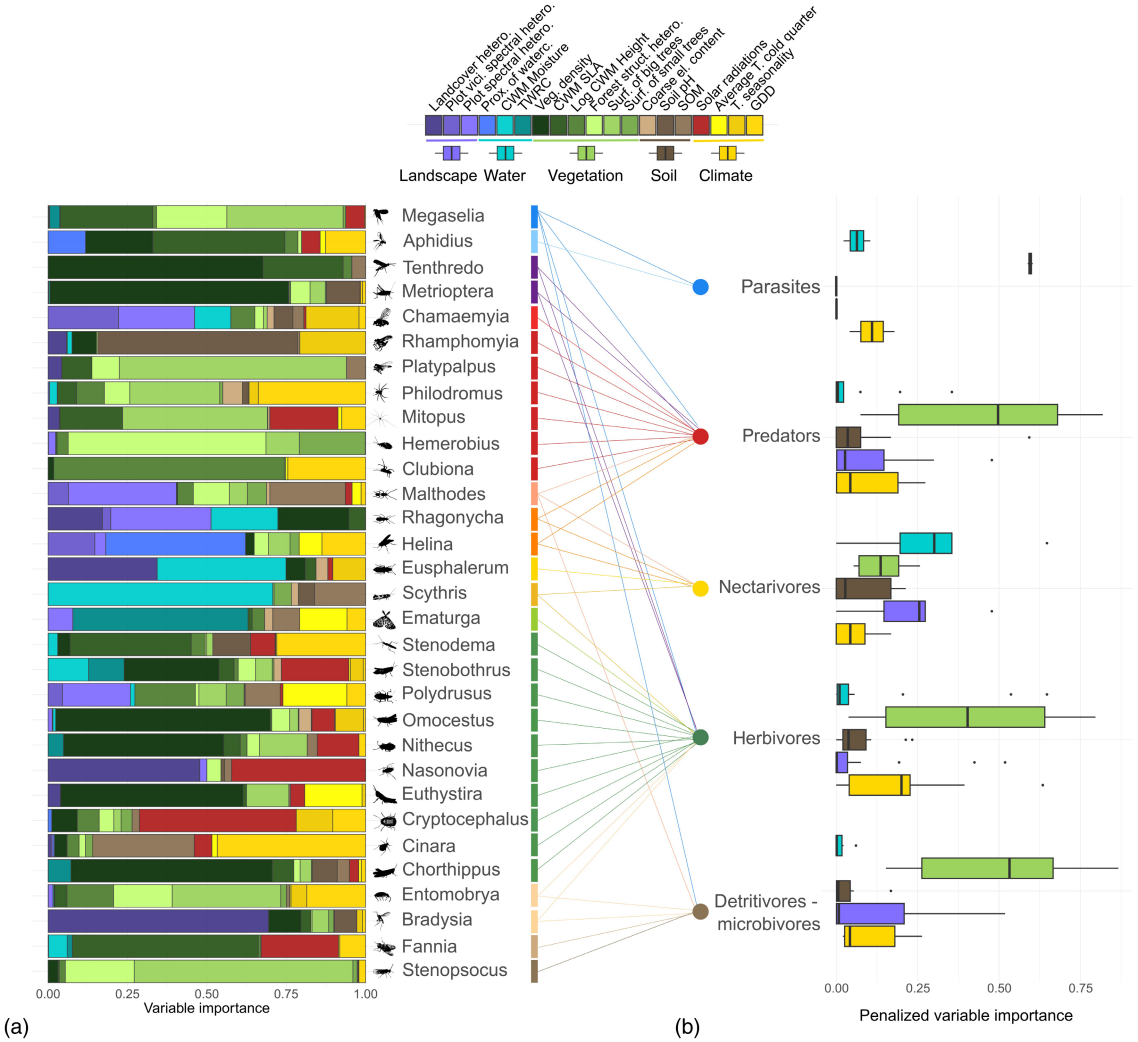



**FIGURE 4** Variable importance across genera and trophic groups of arthropods. (a) Relative variable importance for each of the 31 well-modelled arthropod genera, exhibiting an average cross-validation TSS higher than 0.50. Trophic groups are indicated by corresponding-coloured strips on the right (colours detailed in Figure 2). (b) Variable importance, adjusted based on the average cross-validation TSS of the models, summarized by trophic group and for clusters of variables delineating landscape structure (purple), water availability (blue), vegetation structure (green), soil physico-chemical properties (brown) and climate (red and yellow). Mixed diets have been aggregated into broader groups as guided by the central arrows.

snags, Paillet et al., 2017), and a more complex 3D structure, thus enabling niche partitioning between different parts and layers of vegetation (e.g. Mitopus, Entomobrya and Philodromus, Figures 4 and 5, Jordana & Greenslade, 2020; Michalko & Pekár, 2015). Moreover, dense forests with large trees buffer climatic extremes and retain more humidity than open-habitat (Renaud et al., 2011), therefore favouring genera sensitive to the variability of microclimatic conditions and drought (e.g. *Megaselia*, Durska, 2002). Forest structure heterogeneity, that is, the presence of multiple strata formed by trees of different ages, also seemed to shape the foliar arthropod distributions (Figure 4). For instance, the probability of presence of *Hemerobius* increased with forest structure heterogeneity, likely because the presence of young regenerating trees shielded by mature ones creates favourable conditions for the presence of its prey, such as aphids and soft-bodied insects (Smith, 1923; Straw et al., 2020). Another reason for a positive effect of forest structure on foliar arthropod distributions could be via an indirect effect of tree diversity, as it has been shown to positively influence foliar arthropods diversity (e.g. herbivores, Novotny et al., 2006). On the other hand, the structure of the herbaceous layer significantly influenced the distributions of several arthropod genera associated with grasslands and shrublands in our study system. Many of these genera were favoured by the density and height of the herbaceous layer (e.g. several *Orthoptera*, *Nithecus* and *Clubiona*, Figures 4a and 5), which was strongly correlated with the horizontal variance of plant heights (Pearson's cor = 0.941, $p < 2.2e\text{-}16$). Recent studies suggested that dense, high and structurally complex vegetation offers a favourable balance between predator refuge and resource availability (Joern & Laws, 2013; Perner et al., 2005).

Our study showed that vegetation structure and climate (particularly GDD) hold generally more importance than the

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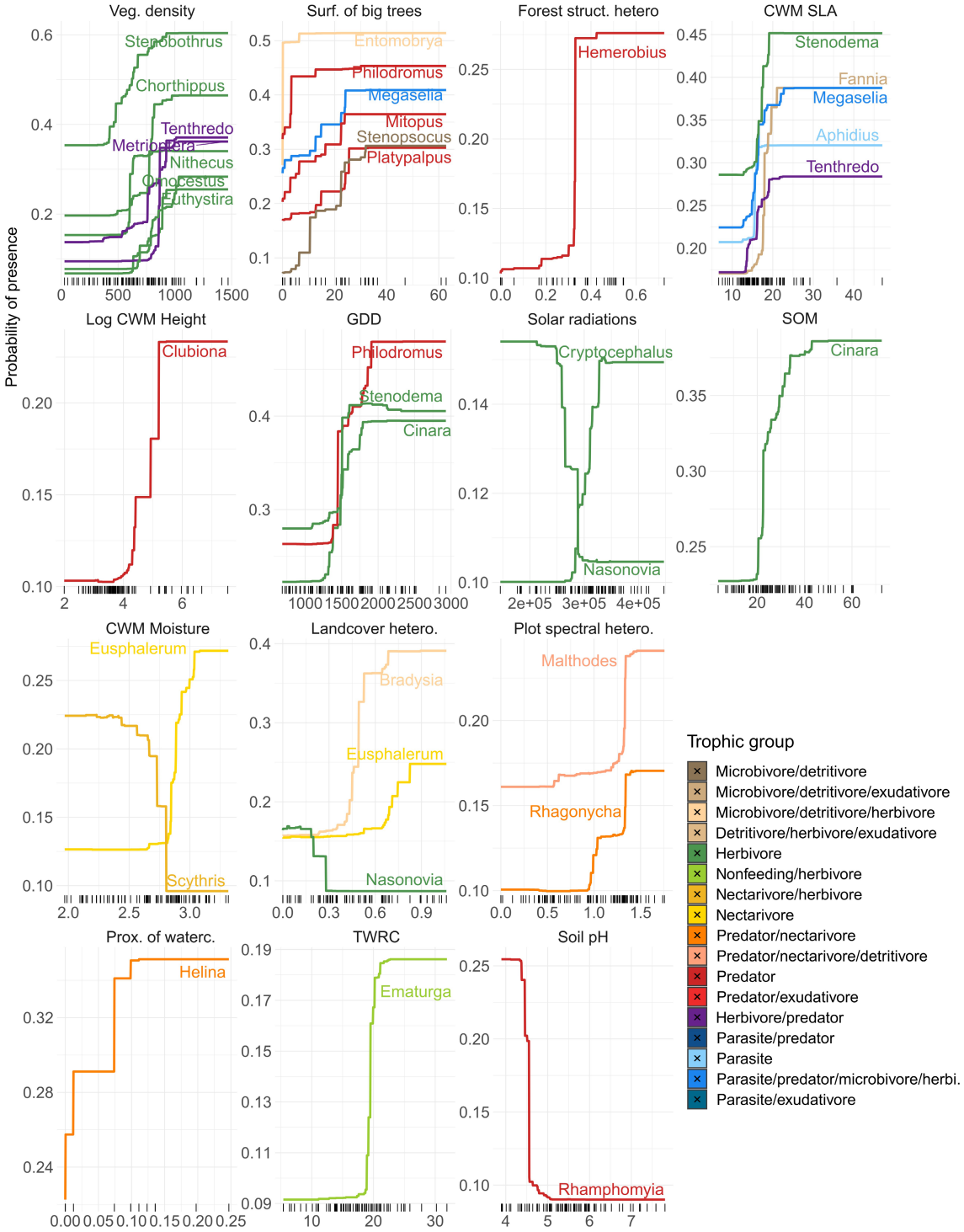



**FIGURE 5** Partial response plots for a selection of genera and predictors. Only the genera for which the relative importance of the predictor higher than 25% are represented. The colours represent the trophic groups. Rug plot along the *x*-axes represents the observed distribution of each environmental predictor.

landscape-related variables (Figures 3 and 4). This contrasts with studies at broader scales (global, continental and national) attributing arthropod decline primarily to landscape structure variables, generally linked to land-use changes and habitat loss (Le Provost et al., 2021; Seibold et al., 2019). Yet, our results align with smaller scale studies emphasizing the role of temperature in mountain areas, where habitats remain relatively preserved. Halsch et al. (2021) proposed that the impact of climate may also be substantial at large scales, potentially rivalling the influence of landscape variables when adjusted for the total loss of natural habitat. More specifically and in line with our results, variables related to mean temperature (e.g. GDD) are frequently considered as more crucial than freezing

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extremes or temperature seasonality. Arthropods success appear to be more closely linked to their optimal body temperature and their capacity to complete development within a given heat budget than to their ability to withstand cold extremes (Hodkinson, 2005; Mani, 2013).

### 4.3 | Consistency of arthropod responses to the environment according to their trophic group

Overall, our findings indicated only subtle variations in the main distribution drivers of distinct trophic groups (Figure 4). But it is worth noting that these different trophic groups may exhibit similar responses to the environment but through varying mechanisms. For example, the positive correlation between the mean specific leaf area (SLA) and the palatability of the herbaceous stratum could explain why high SLA vegetation benefits herbivorous genera (e.g. Stenodema, Gardarin et al., 2014), whereas its association with extrafloral nectaries might elucidate the positive response of exudativores (e.g. *Fannia*, Figure 5, Pereira et al., 2020). Furthermore, the positive effect of mean SLA on lower trophic levels could cascade to higher trophic levels, by fostering increased prey abundance and diversity and thereby promoting the presence of specific predators (e.g. *Tenthredo*) and parasites (e.g. *Aphidius*, Figure 5). Future arthropod distribution models stand to benefit from direct integration of trophic resources as predictor (e.g. diversity of prey for predators, Moreira et al., 2016; Perner et al., 2005; Poggiato et al., 2023), potentially unearthing more pronounced differentials in the responses of trophic groups.

Notably, our models unveiled distinct patterns in the response of nectarivorous genera distributions, which depended more on water availability and landscape heterogeneity than other groups (Figure 4b). This aligns with recent research indicating a strong dependence of nectar production, flower size and quantity on water availability, thereby influencing the attractiveness of plant communities to nectarivores (Kuppler et al., 2021; Kuppler & Kotowska, 2021). Moreover, within mountainous terrain, the likelihood of encountering a patch of nectar-rich flowers likely increases with small-scale spatial diversity. Finally, our results suggest that conservation measures should be adapted to the specificity of nectarivores. In fact, nectarivores play a crucial role in ecosystems, as pollination results from their foraging behaviour, with different efficiencies depending on the nectarivore's traits and taxonomy, its habitat or the plant species in question (McCabe & Cobb, 2021; Wardhaugh, 2015).

## 5 | CONCLUSIONS

In conclusion, using community DNA metabarcoding, our results show the significant importance of vegetation structure and climate in the distribution of foliar arthropods. These results are consistent for all trophic groups, with the exception of nectarivores, whose distributions are more influenced by the presence of water and landscape structure. This study paves the way for multi-trophic studies of foliar arthropod distributions, and indicates that conservation measures should be adapted to the trophic group in question to protect crucial ecosystem functions.

## AUTHOR CONTRIBUTIONS

Camille Martinez-Almoyna (CMA) and Wilfried Thuiller (WT) conceived the ideas and methodology with contribution from Christian Miquel and Frédéric Boyer for metabarcoding laboratory work; CMA, Amélie Saillard (AS), Paul Dufour, Marc Ohlmann (MO), Lily Dunyach, Ruth Stephan and Mary Varoux collected the data; Ludovic Gielly (LG) and CMA carried out the laboratory work for eDNA extractions, LG for PCRs and AS for soil physicochemistry; Clément Lionnet and CMA run the bioinformatic pipelines with contribution from MO; CMA, Sara Si-Moussi, Jérôme Poulenard and Julien Renaud computed the environmental predictors; CMA run all statistical analyses; CMA led the writing of the manuscript together with Irene Calderòn-Sanou, WT and Tamara Münkemüller. All authors contributed to editing and gave final approval for publication.

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## CONFLICT OF INTEREST STATEMENT

The authors have no conflicts of interest to declare.

## DATA AVAILABILITY STATEMENT

Data available from the Dryad Digital Repository: https://doi.org/10.5061/dryad.wdbrv15xr (Martinez-Almoyna et al., 2024).

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## SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

**Appendix 1.** ORCHAMP: Spatio-temporal observatory of biodiversity and ecosystem functioning of mountains' socio-ecosystems.

**Appendix 2.** Supplementary information on sampling dates.

**Appendix 3.** Description of extraction, amplification and sequencing of community DNA metabarcoding of the bulk samples.

**Appendix 4.** Supplementary information on bioinformatic curation pipeline.

**Appendix 5.** Supplementary information on the assignment of genera to trophic groups.

**Appendix 6.** Analyses of diversity alpha and beta at the community level.

**Appendix 7.** Details on environmental data.

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